

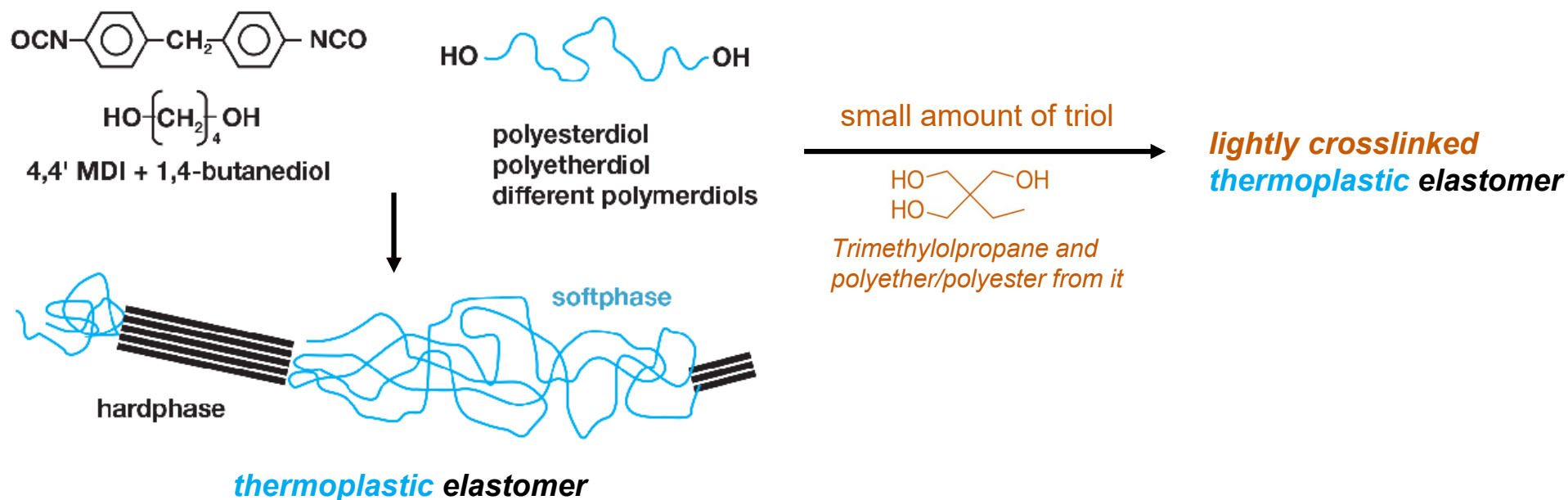
Thermosets

[Lightly Crosslinked Thermosets] → Elastomers

Cast Polyurethane Elastomers (PUE): The Borderline Case

Liquid formulations of diisocyanates (MDI/TDI/IPDI) + polyol (PTHF, PPG) + chain extender (1,4-butanediol, EG)

Microphase-separated hard segments



https://www.tpupolymer.com/what-is-tpu-material_n7

Thermosets

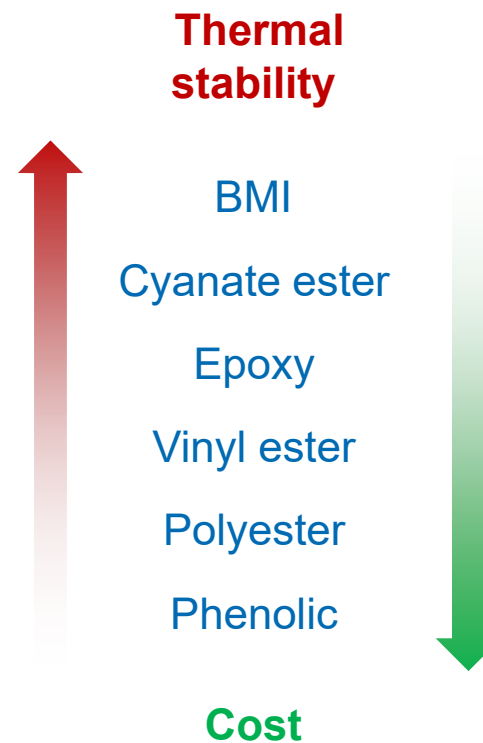
What when?

There Is No "Best" Thermoset!

Performance ↔ Trade-offs

- Cost
- Processing
- Properties
- Lifetime

The Thermoset Selection Map



Thermosets

What when?

A Possible Final Decision Matrix

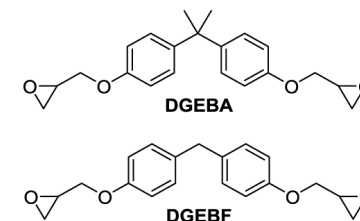
Requirement	Preferred Thermoset
Lowest cost	Unsaturated polyester
Best all-round performance	Epoxy
Flexibility / elastomer	Polyurethane
Fire resistance	Phenolic
Chemical resistance	Vinyl ester
Extreme temperature	BMI
Low dielectric constant	Cyanate ester
Low shrinkage	Benzoxazine

Thermosets

Adhesives & Curing (I)

2K Epoxy Adhesives

Resin and hardeners are mixed just prior to use.



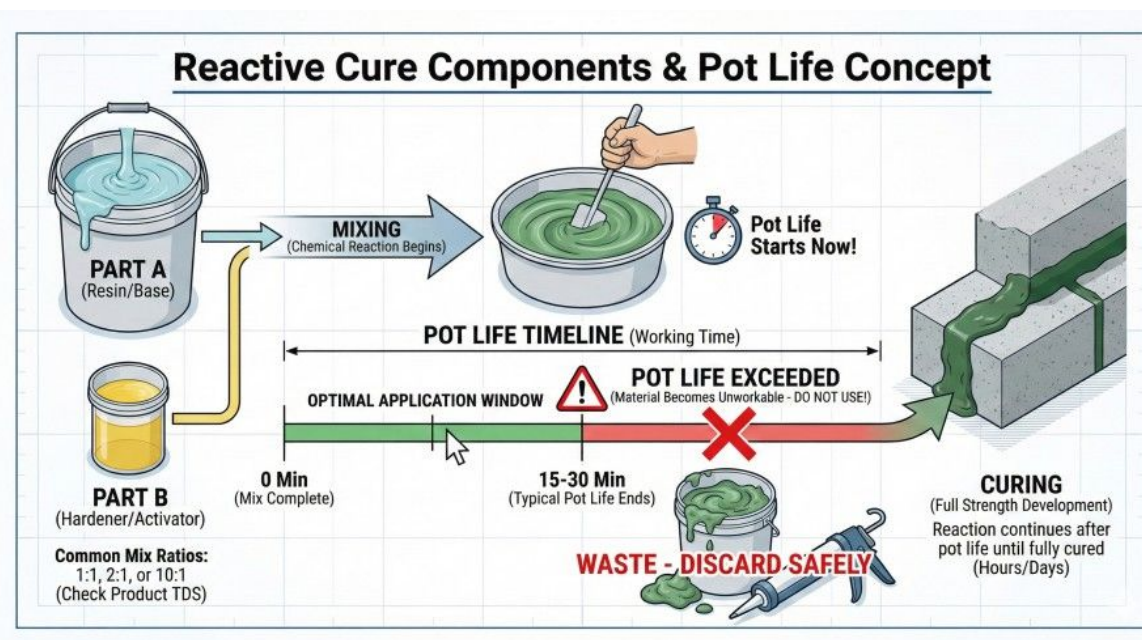
Common formulation:

▪ PART A

- + **epoxy resin** (DGEBA or DGEBF)
- + **tougheners** (elastomer/rubber to prevent brittleness)
- + **fillers & rheology modifiers** (fumed silica and calcium carbonate)

▪ PART B

- + **curing agent** (polyamines, e.g., DETA, polymercaptans)
- + **accelerators** (tertiary amines)



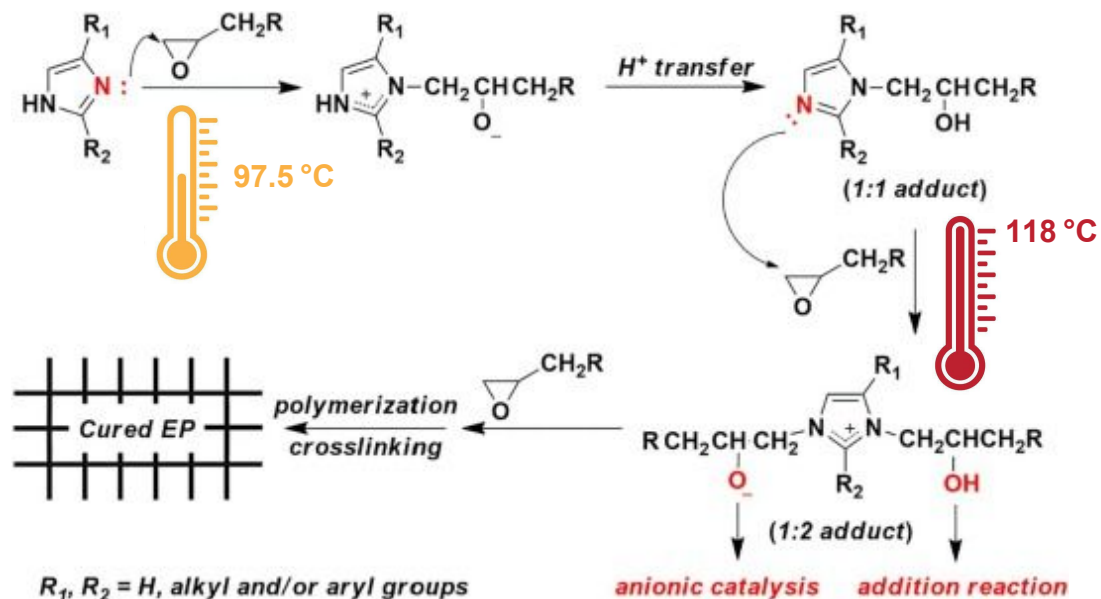
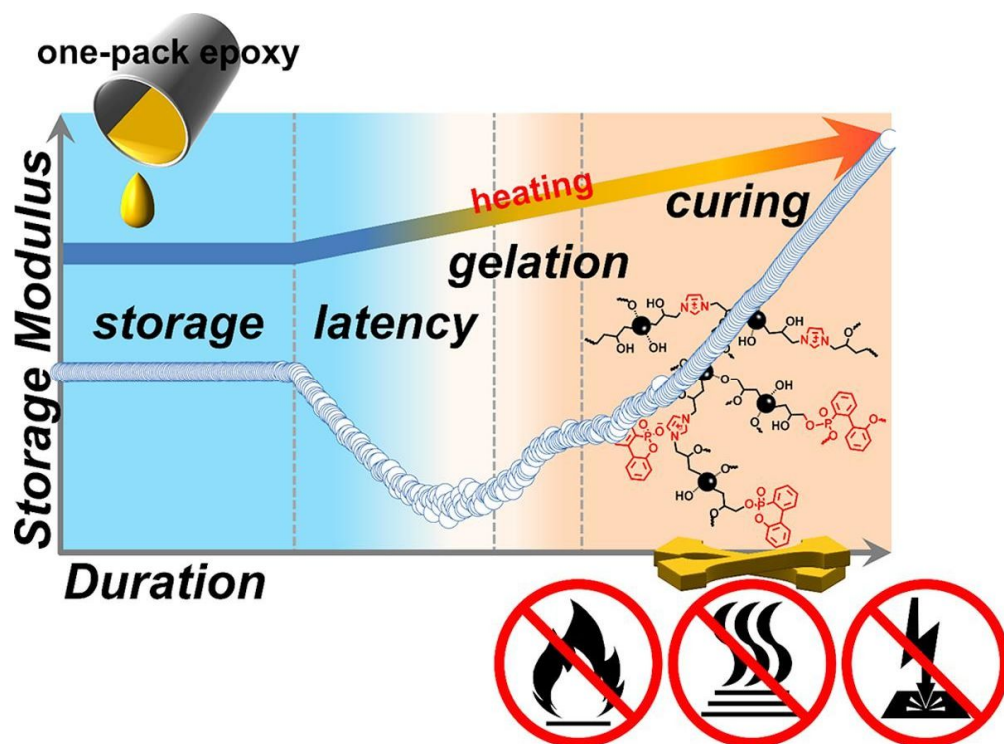
<https://bopinchem.com/vi/sealant-curing-methods-explained-moisture-cure-reactive-uv-cure-and-more/>

Thermosets

Adhesives & Curing (II)

1K Heat-Cured Epoxy Adhesives

Latent hardeners are included in the resin formulation.



Chem. Eng. J. 2018, 347, 223

Thermosets

Adhesives & Curing (III)

1K Heat-Cured Epoxy Adhesives

Common formulation:

- **epoxy resin** (40–70%; DGEBA or Novolac resins).
- **latent hardener** (2–10%; micronized dicyandiamide (DICY), which remains solid at room temperature but melts and cures upon heating)
- **accelerator** (0.1–2%; substituted ureas to lower the curing temperature of the DICY).
- **tougheners** (5–20%; core-shell rubber particles or rubber-modified epoxy prepolymers to prevent the cured joint from becoming brittle)
- **fillers & rheological additives** (10–30%; fumed silica and calcium carbonate)

Thermosets

Adhesives & Curing (IV)

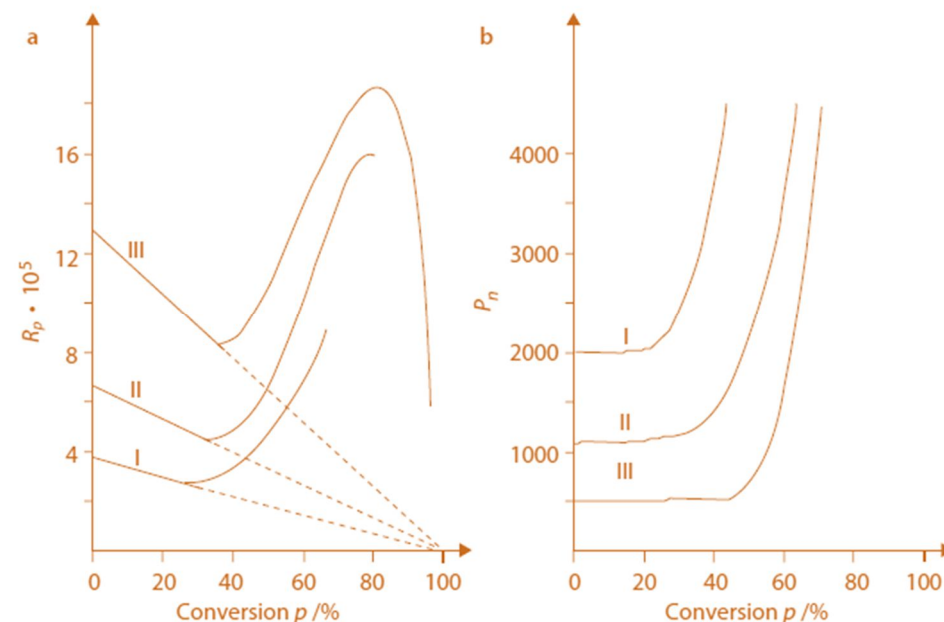
Radical Polymerization Systems

Cure kinetics often display **autoacceleration** (**Trommsdorff effect**) followed by diffusion-controlled slowdown.

As T_g rises during cure, vitrification can arrest conversion and freeze in defects.

⚠ vitrification (mobility) $\neq$ gelation (connectivity)

Post-cure raises T_g and modulus but may not fully heal heterogeneity.

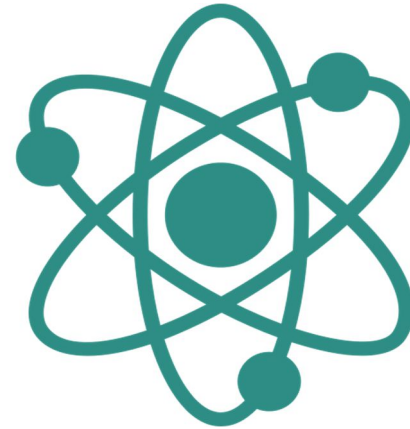


Thermosets

Characterization

▪ Chemistry

- + FTIR/NIR → conversion
- + DSC → monitoring of the curing process (exotherm peaks)
- + solid-state NMR → crosslink motifs
- + extraction process → sol and gel fractions
- + swelling experiment → crosslinking density (compared to DMA → defects)

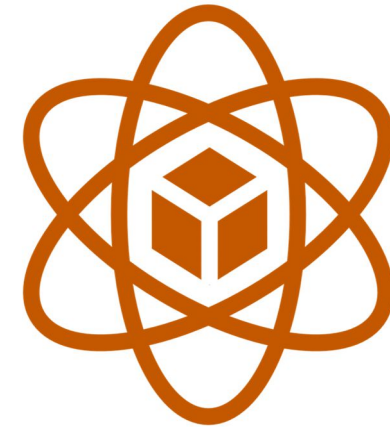


Thermosets

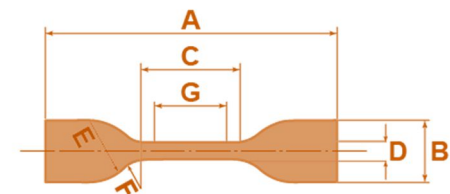
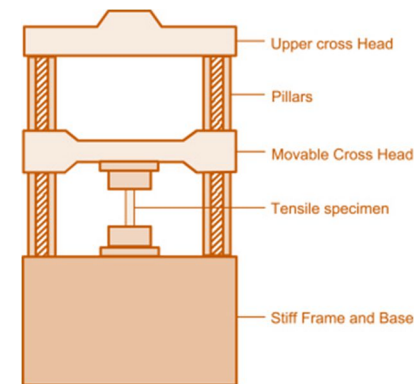
Characterization

Materials Properties

- + DSC $\rightarrow T_g$ of cured materials but also
- + TGA $\rightarrow$ stability and char yield
- + DMA $\rightarrow G'$, G'' , $\tan \delta$, and T_g (rubbery $G \rightarrow \nu$)
- + †tensile testing (ASTM D412) $\rightarrow$ modulus, strength, and elongation
- + ‡tear (ASTM D624) captures crack resistance
- + *durometry $\rightarrow$ Shore X hardness (ASTM D2240) benchmarks feel
- + microscopy $\rightarrow$ structure and morpholog

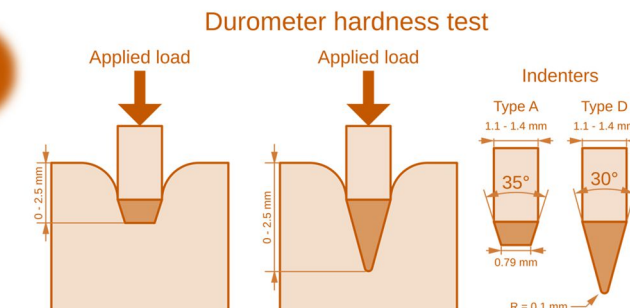


*Encyclopedia of Materials:
Composites 2021, 2, 158*



<https://www.mecmesin.com/standard/iso-37>

<https://www.elastocon.com/learn-more/about-testing/determination-of-tear-strength-of-rubber>



https://en.wikipedia.org/wiki/Shore_durometer

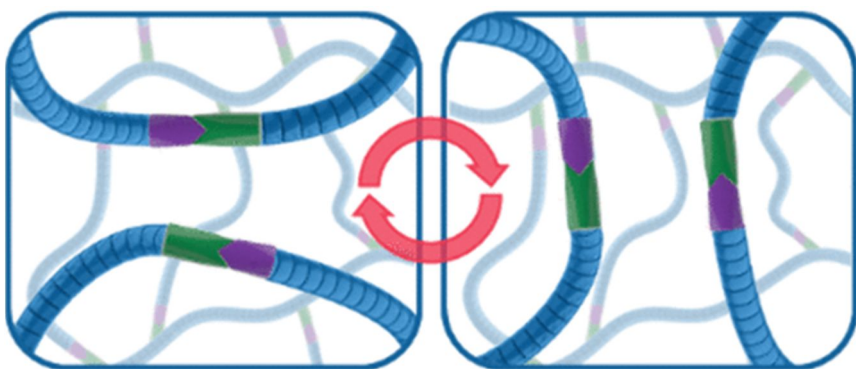
Crosslinked Polymers

Covalent Adaptable Networks

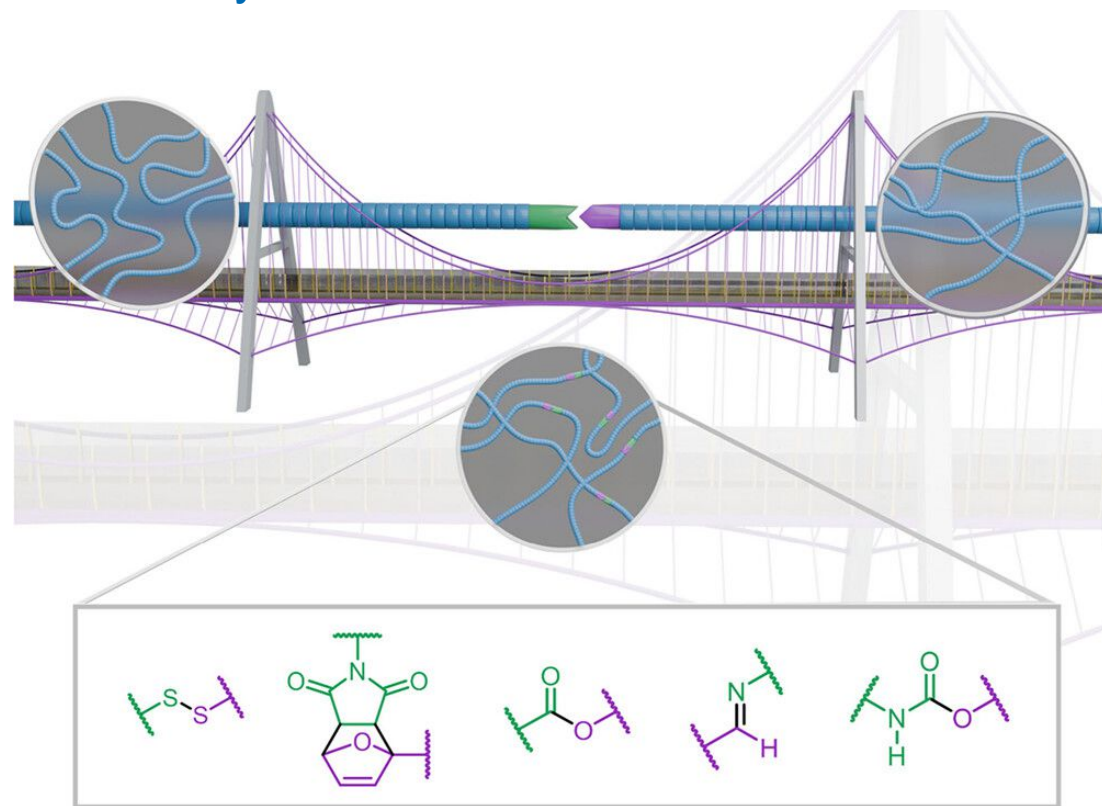
Covalent Adaptable Networks

Generalities (I)

Main Concept: CANs = Reprocessable Crosslinked Polymers



- Topology exchange
- Reprocessability
- Debonding (on demand)

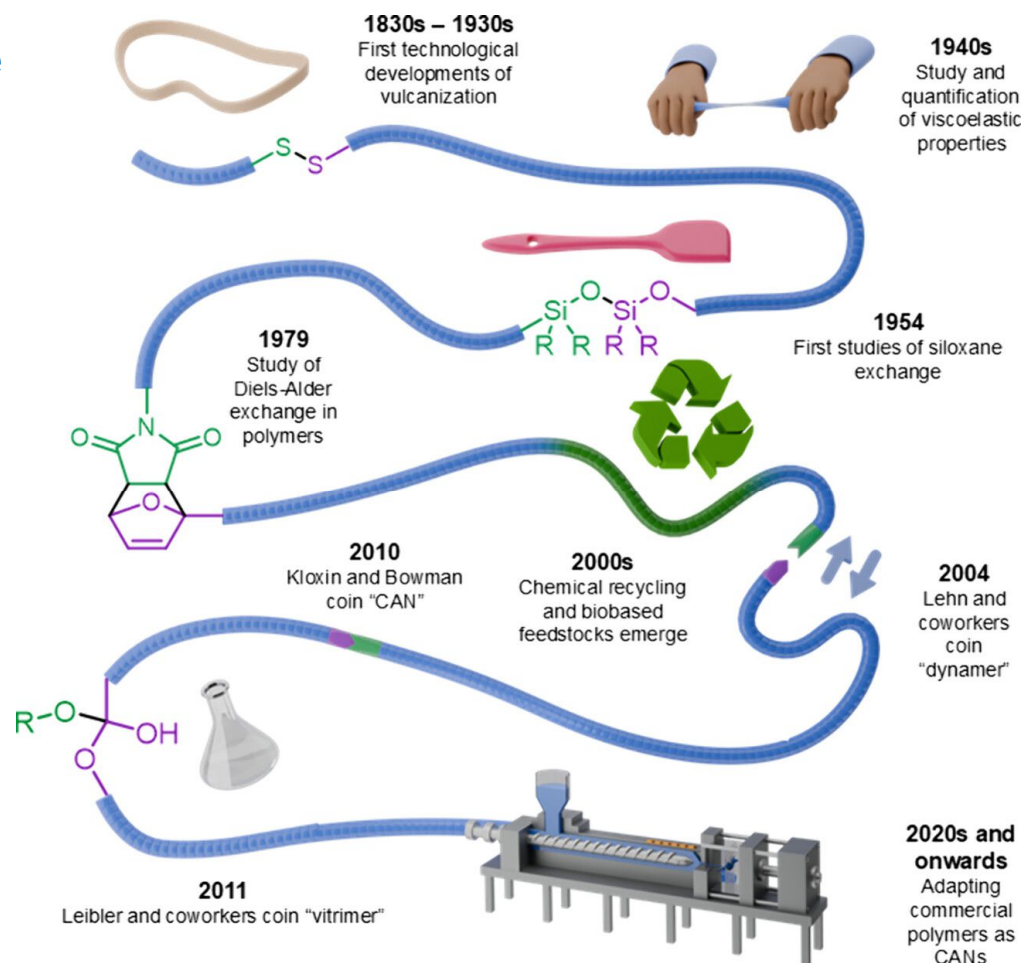


Chem Rev. **2022**, *13*, 370

Covalent Adaptable Networks

Generalities (II)

Historical Perspective



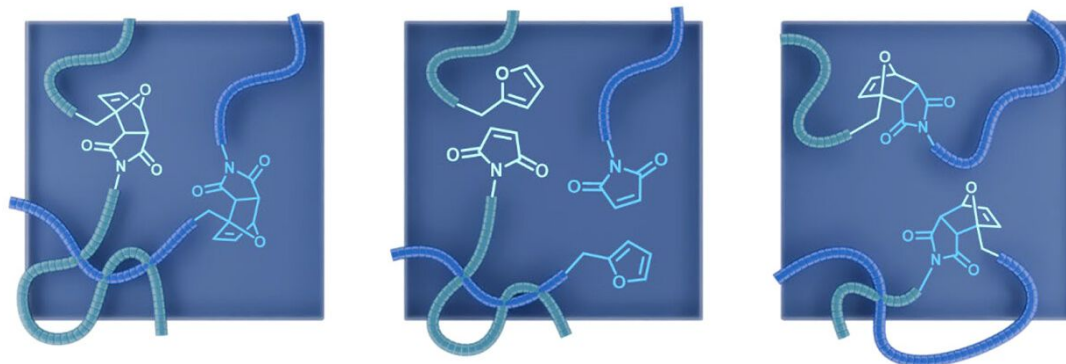
Chem Rev. **2022**, *13*, 370

Covalent Adaptable Networks

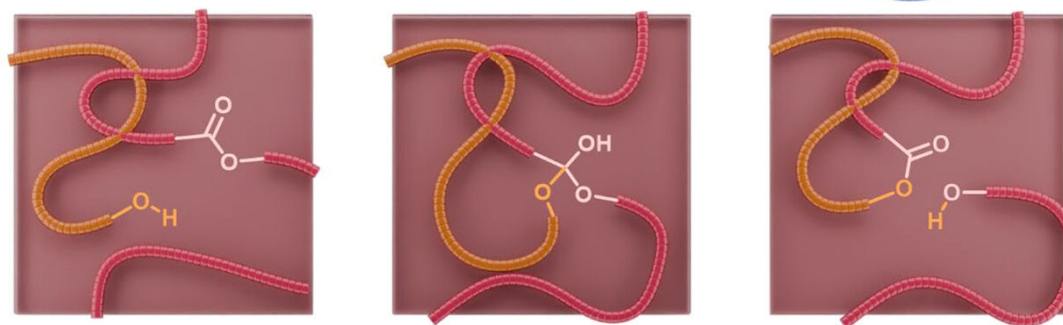
Generalities (III)

Two types of exchange mechanism

- dissociative



- associative

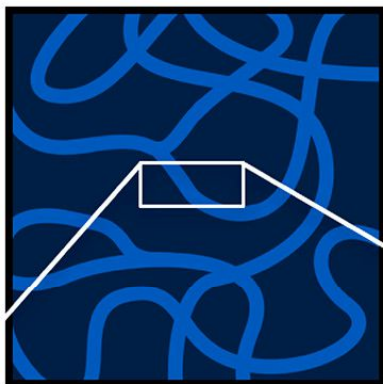


Covalent Adaptable Networks

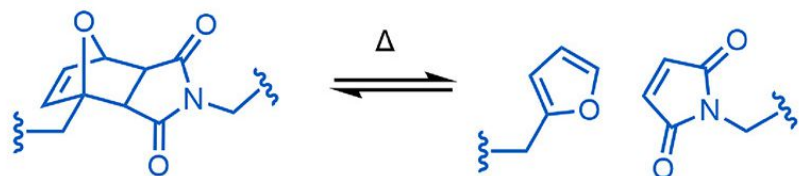
Generalities (III)

Two types of exchange mechanism

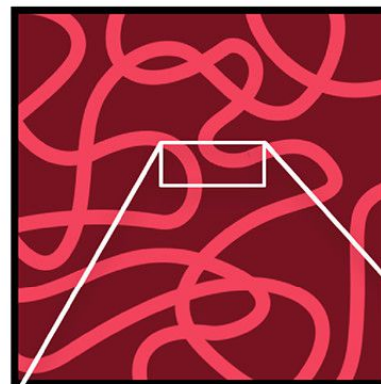
A. Dissociative CANs



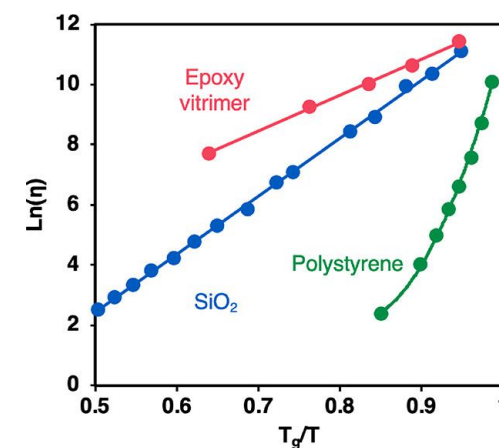
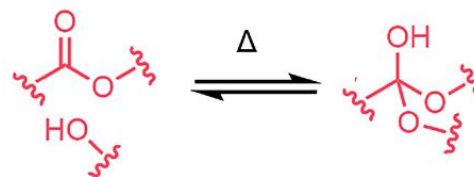
- Bond dissociation precedes bond reformation
- Loss of crosslink density upon heating



B. Associative CANs (Vitrimers)



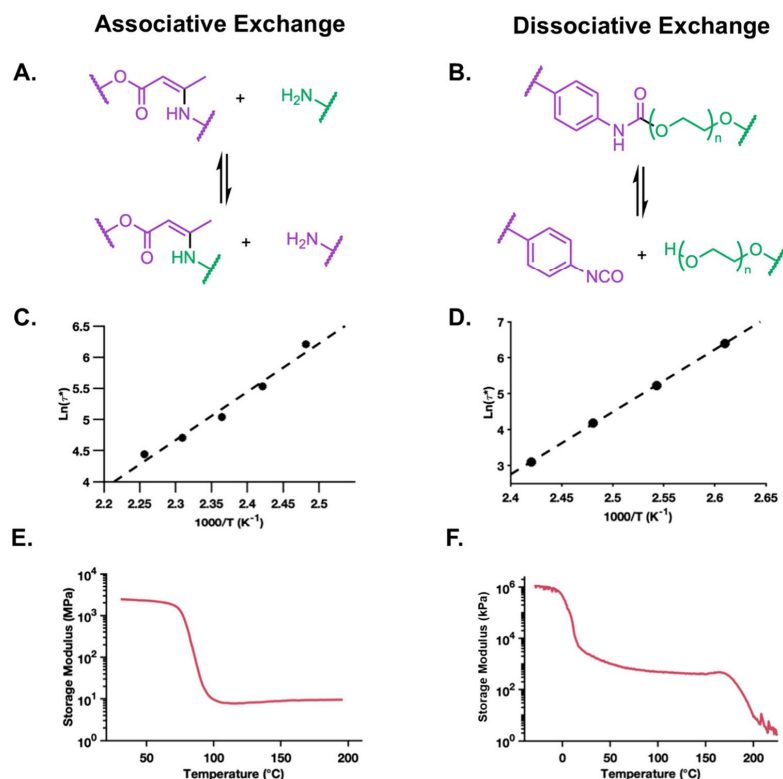
- Bond formation precedes bond dissociation
- Maintain crosslinking density upon heating



Covalent Adaptable Networks

Overview (III)

Two types of exchange mechanism



ACS Central Science > Vol 6/Issue 9 > Article

Open Access

OUTLOOK | July 29, 2020

Reprocessable Cross-Linked Polymer Networks: Are Associative Exchange Mechanisms Desirable? [Click to copy article link](#)

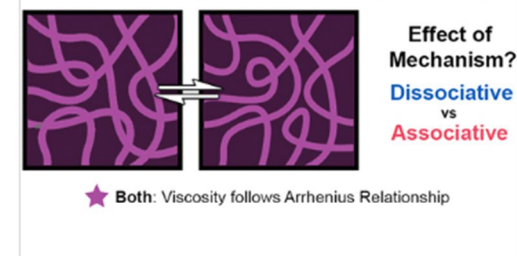
Benjamin R. Elling, and William R. Dichtel*

Open PDF

Abstract

Covalent adaptable networks (CANs) are covalently cross-linked polymers that may be reshaped via cross-linking and/or strand exchange at elevated temperatures. They represent an exciting and rapidly developing frontier in polymer science for their potential as stimuli-responsive materials and to make traditionally nonrecyclable thermosets more sustainable. CANs whose cross-links undergo exchange via associative intermediates rather than dissociating to separate reactive groups are termed vitrimers. Vitrimers were postulated to be an attractive subset of CANs, because associative cross-link exchange mechanisms maintain the original cross-link density of the network throughout the exchange process. As a result, associative CANs demonstrate a gradual, Arrhenius-like reduction in viscosity at elevated temperatures while maintaining mechanical integrity. In contrast, CANs reprocessed by dissociation and reformation of cross-links have been postulated to exhibit a more rapid decrease in viscosity with increasing temperature. Here, we survey the stress relaxation behavior of all dissociative CANs for which variable temperature stress relaxation or viscosity data are reported to date. All exhibit an Arrhenius relationship between temperature and viscosity, as only a small percentage of the cross-links are broken instantaneously under typical reprocessing conditions. As such, dissociative and associative CANs show nearly identical reprocessing behavior over broad temperature ranges typically used for reprocessing. Given that the term vitrimer was coined to highlight an Arrhenius relationship between viscosity and temperature, in analogy to vitreous glasses, we discourage its continued use to describe associative CANs. The realization that the cross-link exchange mechanism does not greatly influence the practical reprocessing behavior of most CANs suggests that exchange chemistries can be considered with fewer constraints, focusing instead on their activation parameters, synthetic convenience, and application-specific considerations.

Covalent Adaptable Network Reprocessing



Chem Rev. 2022, 13, 370
ACS Cent. Sci. 2020, 6, 1488

Covalent Adaptable Networks

Overview (III)

Two types of exchange mechanism

It may also be relevant to consider the extent of exchange, particularly for dissociative exchange:

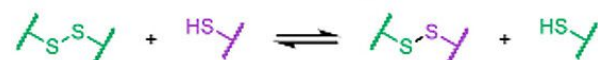
- partial exchange (the material is still crosslinked and infusible);
- (near-)quantitative (the material can molecularly dissolve).

Covalent Adaptable Networks

Disulfide Exchange (I)

The mechanism of exchange may vary based on different stimuli, including temperature, light, reducing/oxidative conditions, or changes in pH.

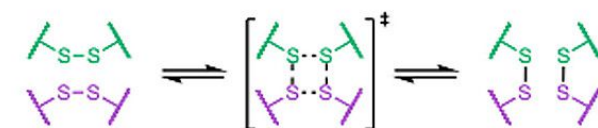
A. Thiol-disulfide exchange



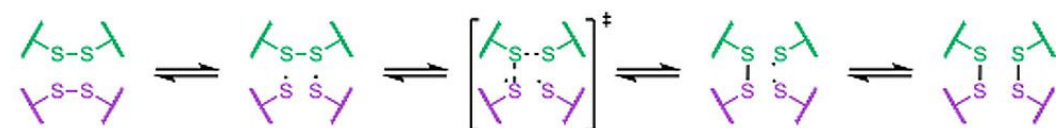
B. Redox disulfide



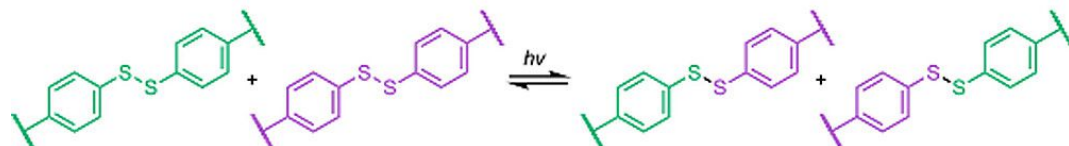
C. Disulfide metathesis



D. [2+1] radical disulfide exchange



E. Light activated radical aromatic radical exchange



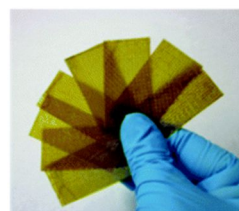
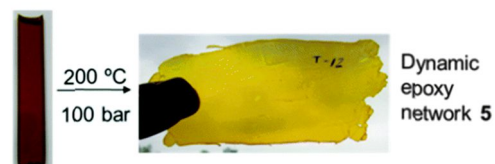
Chem Rev. **2022**, *13*, 370

Covalent Adaptable Networks

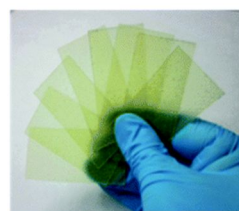
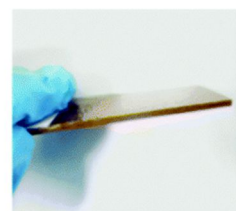
Disulfide Exchange (II)

Example: *Epoxy resin*

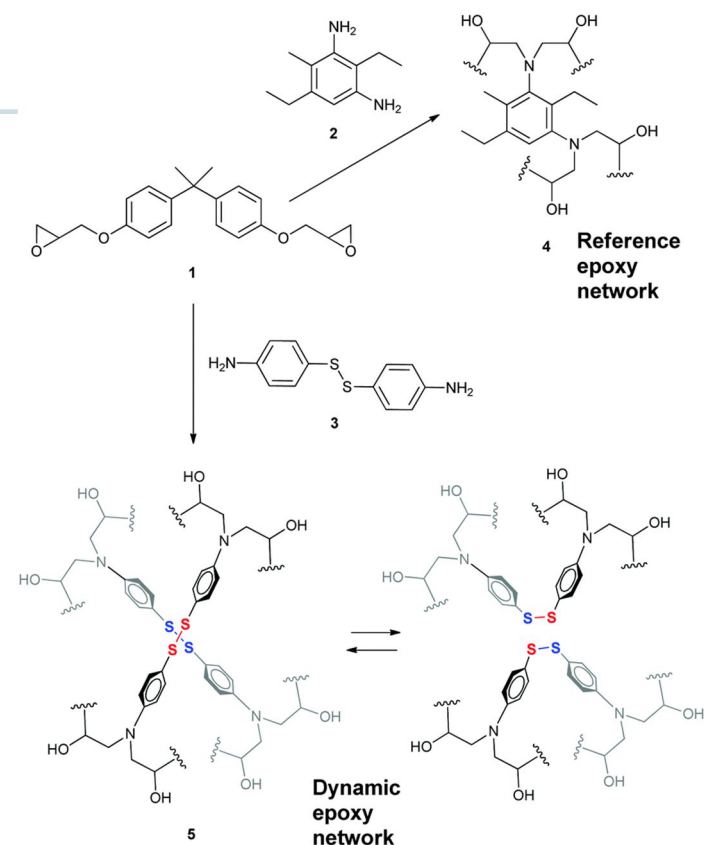
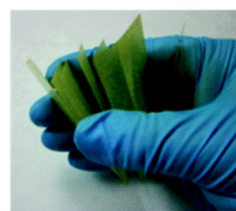
with exchangeable **disulfide** crosslinks



200 °C
100 bar



200 °C
100 bar



	T_g (DSC) [°C]	T_g (DMA) [°C]	T_d [°C]	E' (25 °C) [GPa]	E' (150 °C) [MPa]	Stress ^a [MPa]	Strain ^a [%]
4	127	130	350	2.5	20	81	7.3
5	130	130	300	2.6	20	88	7.1

^a Values at break.

Mater. Horiz. **2016**, *3*, 241

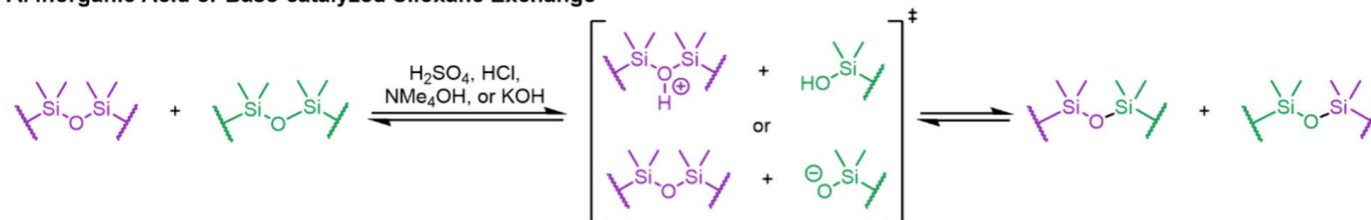
Covalent Adaptable Networks

Siloxane and Silyl Ether Exchange (I)

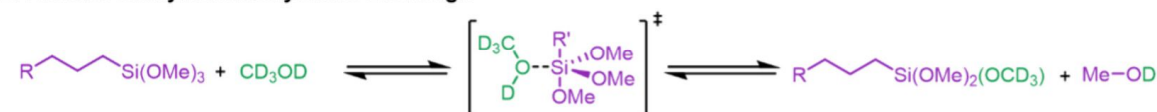
–Si–O– exchange

Acid/base catalysis

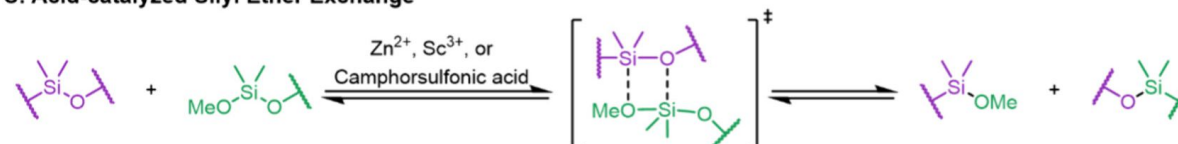
A. Inorganic Acid or Base-catalyzed Siloxane Exchange



B. Alcohol-catalyzed Alkoxysilane Exchange



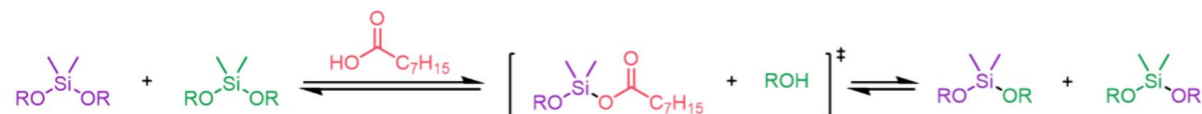
C. Acid-catalyzed Silyl Ether Exchange



D. Base-catalyzed Silanol-mediated Siloxane Exchange



E. Octanoic Acid-catalyzed Silyl Ether Exchange



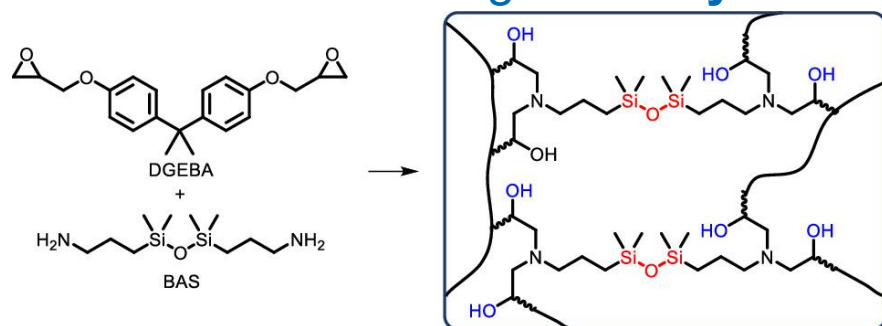
Chem Rev. **2022**, *13*, 370

Covalent Adaptable Networks

Silyl Ether Exchange (II)

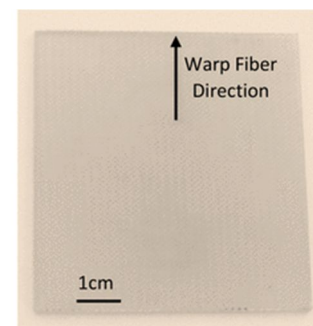
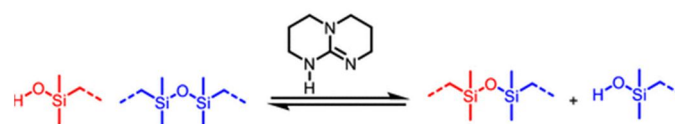
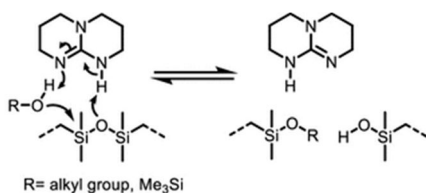
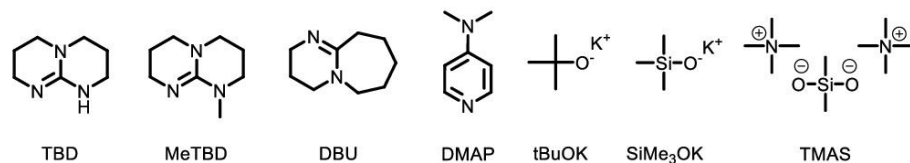
Example: *Epoxy resin*

with exchangeable **silyl ether** crosslinks

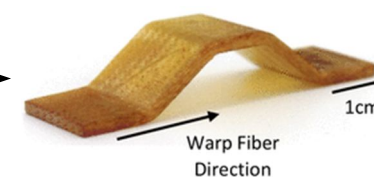


entry a	T_g^{DSC} (°C)	T_g^{DMA} (°C)	soluble fraction b (wt %)	swelling degree c (%)	τ^* (s)
1 (R0)	85	85	1	15	~6
2 (R1)	85	81	1	- d	~6
3 (R2)	85	83	1	11	~6

Catalysts



Thermoforming
190°C, 20 min
30 bar

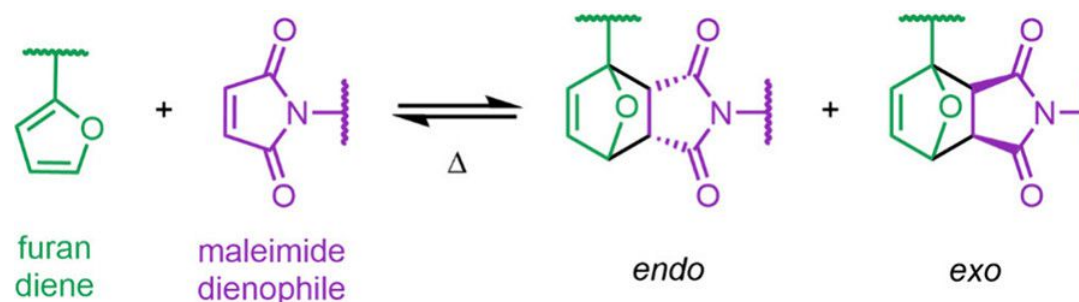


J. Am. Chem. Soc. **2022**, *144*, 12280

Covalent Adaptable Networks

Diels–Alder Adducts (I)

Thermoreversible cycloaddition, most prominently with furan and maleimide derivatives

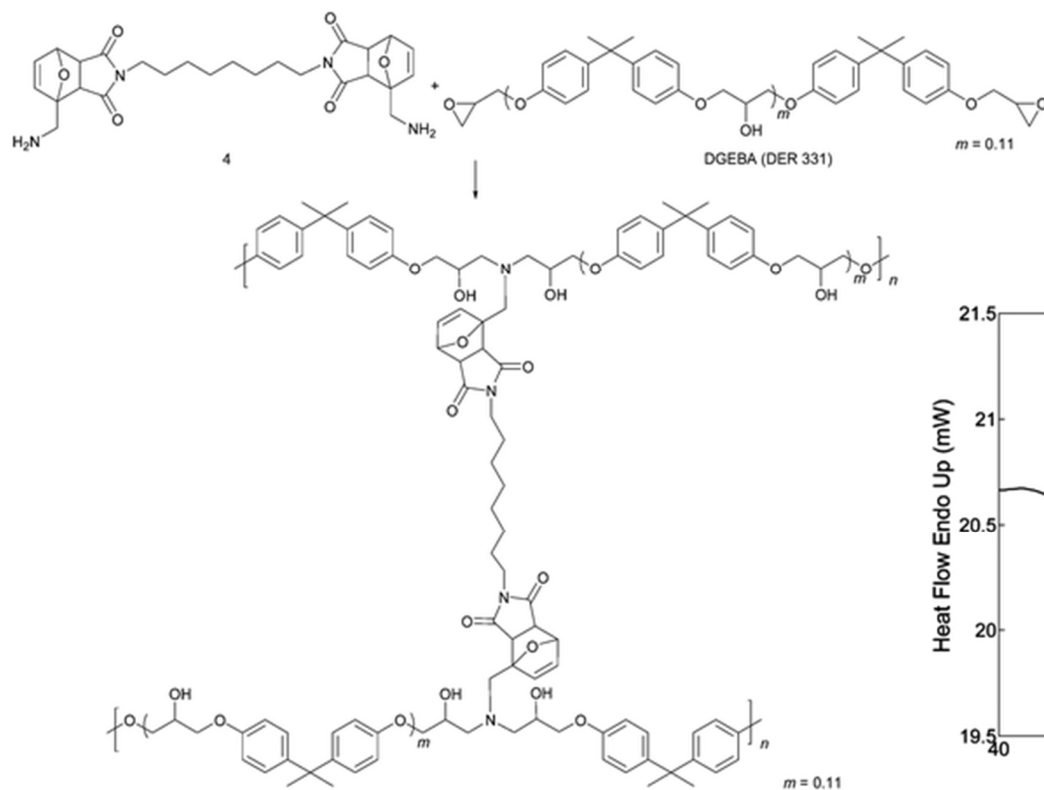


Covalent Adaptable Networks

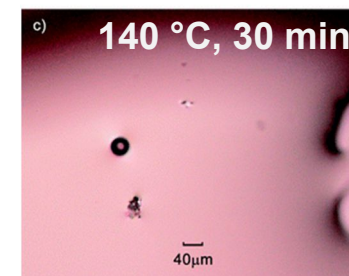
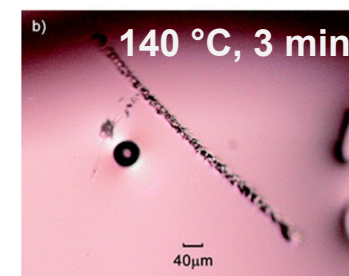
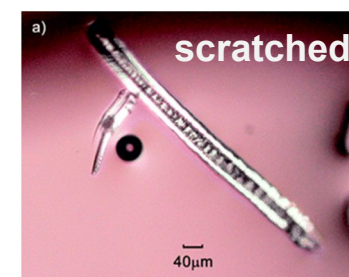
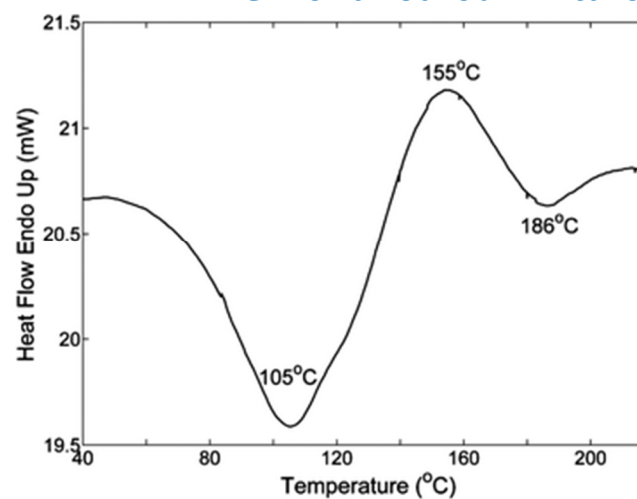
Diels–Alder Adducts (II)

Example: *Epoxy resin*

with exchangeable **Diels–Alder adduct** crosslinks



DSC of uncured mixture



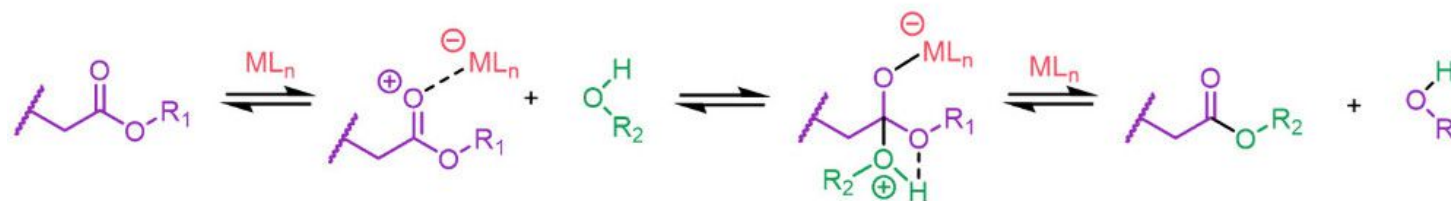
Polym. Chem. **2013**, *4*, 724

Covalent Adaptable Networks

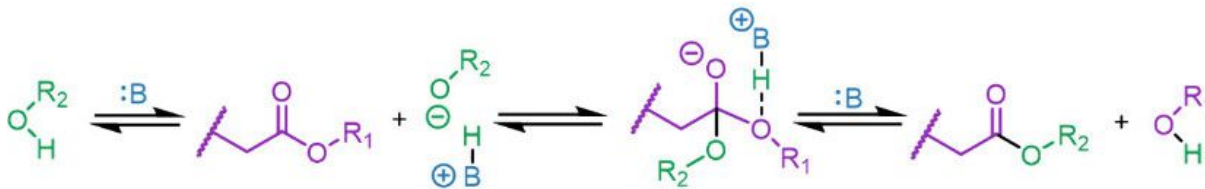
Transesterification Vitrimers (I)

Zn organocatalyst

A. Acid-catalyzed transesterification



B. Base-catalyzed transesterification

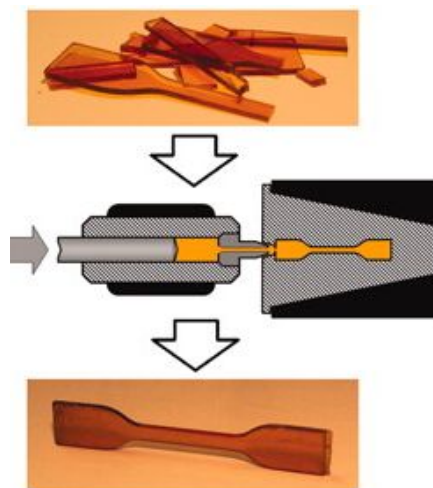
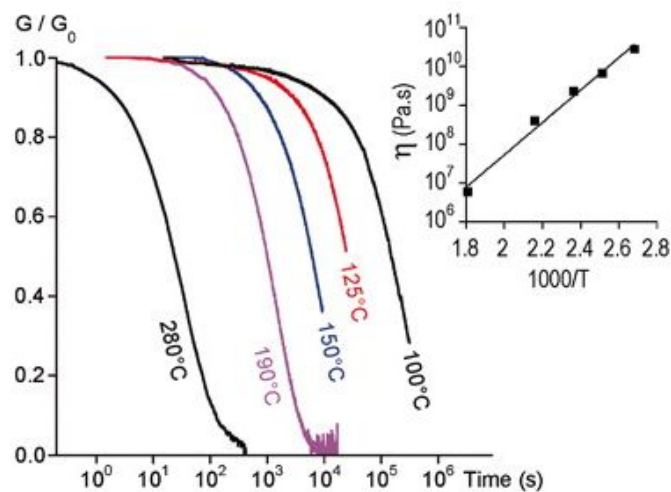
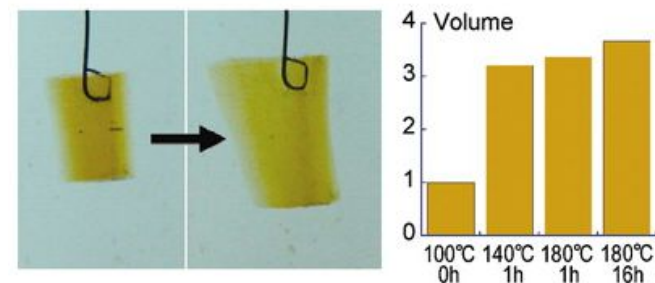
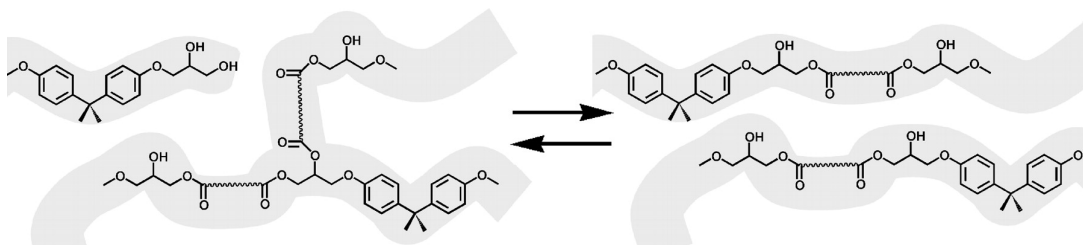


Covalent Adaptable Networks

Transesterification Vitrimers (I)

Example: *Epoxy resin*

with exchangeable **ester** crosslinks (Zn organocatalysis)



Science **2022**, 13, 370

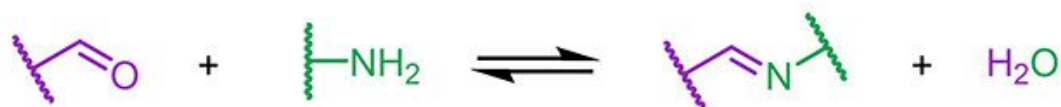
Covalent Adaptable Networks

Imine/Oxime Transimination (I)

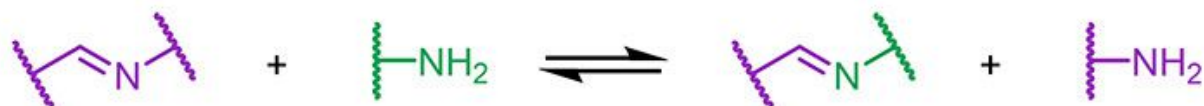
pH/moisture-driven

Mild conditions

A. Condensation and Hydrolysis



B. Transimination



C. Imine Metathesis



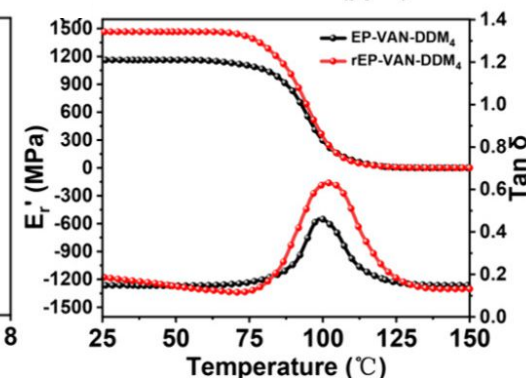
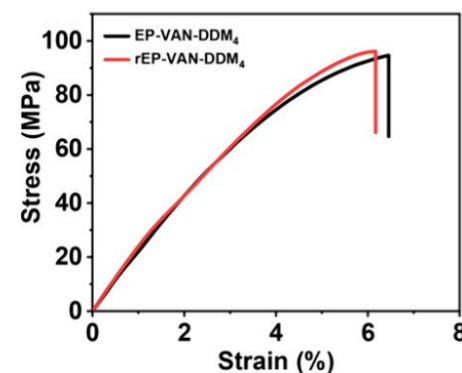
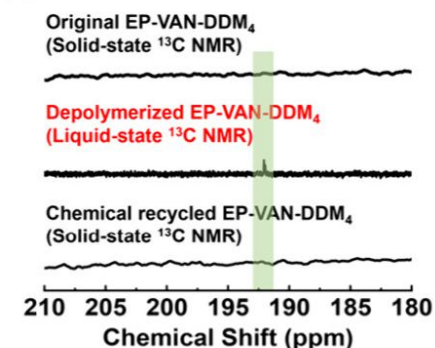
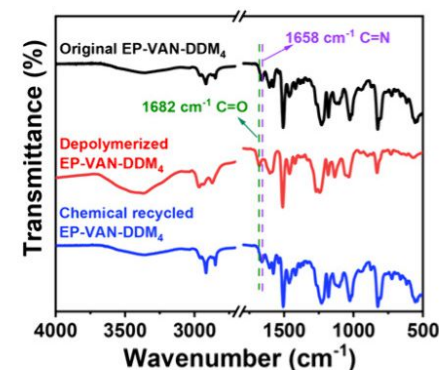
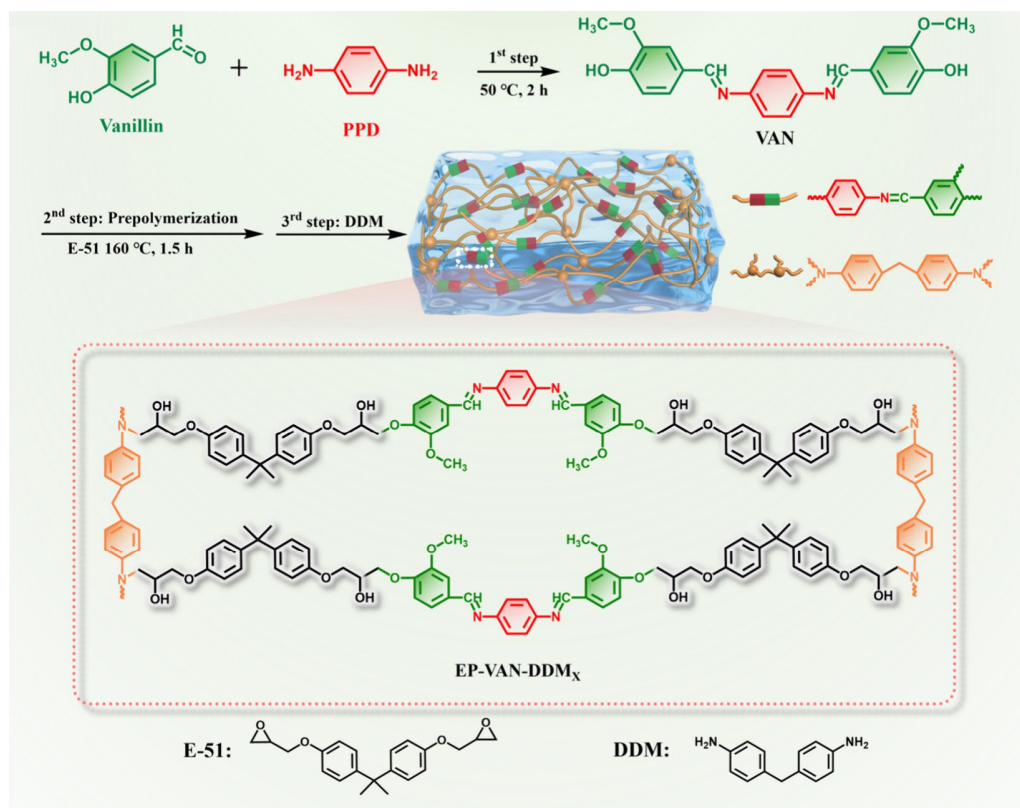
Chem Rev. **2022**, *13*, 370

Covalent Adaptable Networks

Imine/Oxime Transimination (II)

Example: *Epoxy resin*

with exchangeable **imine** crosslinks



Science **2022**, 13, 370